Date: / /

MTS Telangana 2020

1 $(1+2) \times (-4) \div 5$

$= 3 \times (-4) \div 5 = -12/5$ Ans

2 $0.5\% = \frac{0.5}{100} = 0.005$ Ans

3 Principal = 5200 Time = 2 years

Rate = 6%

$SI = \frac{P \times R \times T}{100} = \frac{5200 \times 2 \times 6}{100}$

$= 52 \times 2 \times 6 = 624$ R.S

4 Walking + Riding = 6.25 hours. [6 hours $\frac{15}{60}$ hours = 6 hour $\frac{1}{4}$ hour = 6.25 hour)

Walking + Walking = 7.75 hours.

⇒ Walking one way = $\frac{7.75}{2}$ = 3.875 hours.

So, Riding one way = 6.25 − 3.875 = 2.375 hours.

⇒ Riding on both ways = 2.375 × 2 = 4.750 hours.

= 4 hours 45 minutes.

5 Ans 15 egs = 75 R.S

⇒ 1 egg = $\frac{75}{15}$ = 5 R.S

Now 4 dozen eggs = 12 × 4 = 48 eggs.

Price = 48 × 5 = 240 R.S.

46 Ans $\frac{2}{5} \div \frac{3}{4} = \frac{2}{5} \times \frac{4}{3} = \frac{8}{15}$ Ans

POSTAL A B C

47 Ans Total income of P and Q = 6000×2 = 12000 R.S. ①

Total income of Q and R = 5250×2 = 10500 R.S ②

Total income of P and R = 5500×2 = 11000 R.S. ③

⇒ By adding ①, ② and ③ we get

$P+Q = 12000$

$Q+R = 10500$

$P+R = 11000$

$2(P+Q+R) = 33500$

$\Rightarrow P+Q+R = \frac{33500}{2} = 16750$ — ④

Subtract ② from ④ we get

$P+Q+R = 16750$

$-\quad Q+R = 10500$

$P = 6250$ Ans

48 Ans Cost price = 8220 R.S

Profit % = 25%

$P\% = \frac{Profit}{CP} \times 100$

$\Rightarrow 25 = \frac{Profit}{8220} \times 100 \Rightarrow P = \frac{25 \times 822}{10} = 2055$ R.S.

Now SP = CP + Profit = 8220 + 2055 = 10275 R.S.

Now IInd condition

C.P = 10275

Loss = $10275 \times \frac{20}{100} = 2055$ Loss.

So No Loss, No gain.

POSTAL A B C

49 Ans ① A + B = 15 days. → 40 U/day
② B = 20 days → 30 U/dy.
60 Unit
Let LCM = Total work.

Subtract ② from ① we get

$$A + B = 4$$
$$-\ B = 3$$
$$A = 1 \text{ Unit/day}$$

⇒ Time taken by A to complete 60 Units of work

$$= \frac{60}{1} = 60 \text{ days.}$$

50 Ans Let number of notes of each denomination = x,

So total notes = x + x + x = 3x.

Now According to question

Value of notes = x + 5x + 10x = 480

⇒ 16x = 480

and $x = \frac{480}{16} = 30$ R.S.

So total notes = 30 × 3 = 90 notes.

POSTAL A B C